Experimental Evaluation of Dynamic-Fee Hooks (draft section for the ICBC paper)

I. Experimental Evaluation

The framework's evaluation layer runs every fee policy and a family of static baselines over identical replayed market histories and scores the liquidity provider (LP) against simply holding the deposited tokens. This section describes the simulated market, the two-flow trader model, the experiment design, and the results.

A. Simulated Market and Reproducibility

The experiment is a factorial matrix; one *cell* of it — one combination of fee policy, trading pair, day-window, and gas-price scenario — replays a *window* of 1,440 one-minute Binance candles from 2024 against a Uniswap v4 pool holding a 20M USDT basket, split equally by value into a full-range position. The pool opens at the external market price of the first candle; a mock Chainlink-compatible feed tracks each candle's close; the first 60 candles execute but are excluded from measurement. Three pairs are studied: ETH/SHIB and ETH/USDC (volatile) and USDC/USDT (stable, with a synthetic unit feed for its second leg). Windows are stratified by realized volatility into terciles — 24 calm, 24 medium, and 24 stormy days per pair — so market regimes are represented by design.

Every cell records the manifest of Eq. (M9)[1]: the configured contract, a hash of the full contract source tree, the initial price and liquidity, a fingerprint of the window's own candles, and the RNG seed. A rerun skips cells whose manifest matches, and any change to a contract or input invalidates exactly the affected cells. Because every swap executes in a real EVM (Foundry), gas is *measured* rather than assumed: a calibration run records each policy's median gas per swap (76k for the static hook, 88–124k for the adaptive ones), and these measured figures feed back into both traders' decisions below. Fees follow v4 accounting exactly: they accrue to the position and do not compound. As an end-to-end validity check, the LP's holdings are reconstructed independently from pool state and from trader flows; the two ledgers agree to a relative error of $\sim 10^{-16}$ on every cell.

B. Two-Flow Trader Model

Following the standard microstructure framing, order flow is split into an informed arbitrageur and uninformed users (UU).

[1]References of the form Eq. (Mn) point to equation n of the *main paper* (`main.tex`), not to this draft's own numbering; they become ordinary `\refs` at merge time. Here: Eq. (M9) is the run identity $E = \langle C_\theta, D, P_0, L_0, W, S\rangle$.

1) Arbitrageur: With fee fraction f and $\gamma = 1-f$, the pool price may sit anywhere inside the no-arbitrage band $[P\gamma,\ P/\gamma]$ around the external price P. Once per candle, after the retail flow, the arbitrageur moves the pool to the nearest band edge if that trade is profitable *after gas*: with pool liquidity L, sqrt-price s and target t, the closed-form profit

$$\Pi^*_{0\to 1} = \frac{L(s-t)^2}{s}, \qquad \Pi^*_{1\to 0} = \frac{L(t-s)^2}{\gamma s}, \tag{1}$$

must exceed $g\,p_{\text{gas}}$, where g is the policy's measured median gas and p_{gas} the scenario's gas price (5, 20, or 80 gwei), valued at the ETH price series. For the size-dependent `MEVChargeHook` the optimum is found by ternary search; profit is unimodal in size. Π^* is a decision quantity only: all reported metrics are computed from the balance changes of executed swaps.

2) Uninformed users: UU behavior follows the acceptance model of our prior work [1]: a candidate trade evaluates the normalized outcome ratio

$$r = \frac{V_{\text{out}} - V_{\text{in}} - G}{V_{\text{out}} + V_{\text{in}} + G}, \tag{2}$$

where V_{in} and V_{out} are the two legs valued at external prices — so fee *and* slippage are inside r — and G is the trade's own gas. Execution is probabilistic,

$$P_{\text{exec}} = \begin{cases} 1, & r \geq 0, \\ e^{-\lambda |r|}, & r < 0, \end{cases} \tag{3}$$

with a sensitivity calibration $\lambda = \ln 2 / |r(30\,\text{bps})| \approx 461.4$: half of the willing flow declines at a 30 bps total transaction cost. (λ is a stated assumption and is varied in sensitivity analysis; at our trade sizes the uncalibrated $e^{-|r|}$ is insensitive to any fee in the box.)

Each minute c, each direction independently draws one candidate trade of size

$$q(c) = \frac{v_0(c)}{2}\, e^{\sigma Z - \sigma^2/2}, \qquad Z \sim \mathcal{N}(0,1), \quad \sigma = 1, \tag{4}$$

where $v_0(c)$ is the minute's potential demand (Eq. (5) below), split equally between the two directions. The lognormal multiplier is mean-preserving ($\mathbb{E}[e^{\sigma Z - \sigma^2/2}] = 1$), so randomness produces realistic heavy-tailed trade sizes without distorting the calibrated volume; the trade then executes all-or-nothing with probability P_{exec}. Discreteness is essential: it displaces the pool (giving the arbitrageur work), gives each trade a definite gas bill, and keeps randomness reproducible — all draws are pre-generated from a recorded seed. One draw per direction per minute is the lumpiest reading of the measured volume and is declared as a limitation.

TABLE I
THE OPERATING POINTS READ AS KINDS OF POOLS. ARBITRAGE SHARES OF EXECUTED VOLUME AS MEASURED AT THE 30 BPS BASELINE.

κ	**demand/day**	**the pool it describes**	**arb**
3.0	60M (300% TVL)	flagship, retail-flooded	16%
1.0	20M (100% TVL)	active primary venue (reference)	13%
0.3	6M (30% TVL)	mid-tier pool	25%
0.1	2M (10% TVL)	secondary venue	58%
0.03	0.6M (3% TVL)	over-provisioned pool	89%

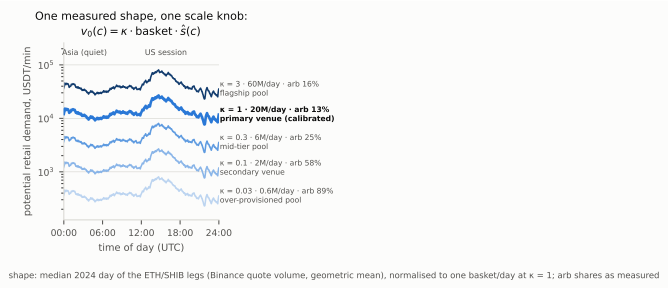

Fig. 1. Demand calibration, Eq. (5): the intraday shape $\hat{s}(c)$ is measured (median 2024 day of the pair's Binance legs); the single knob κ scales its height, and the height sets the arbitrage share of volume the experiment measures.

3) Demand calibration: The *shape* of retail demand is measured: with $\hat{s}(c)$ the 2024 intraday profile of the pair's constituent Binance markets (geometric mean of the two legs' per-minute quote volume, normalized to sum to one over an average day), the potential demand of minute c is

$$v_0(c) = \kappa \cdot \text{basket} \cdot \hat{s}(c), \tag{5}$$

so the single free parameter κ is the pool's daily turnover in baskets per day. The reference point $\kappa = 1$ — unit turnover by the rule above, not a fit to real pools — describes an active primary-venue pool. It is singled out for one reason only: as the unit of the parameterization it is the sole level with an a-priori claim, fixed before any result existed, so the headline comparison cannot have been chosen to flatter or condemn any policy; every finding is then checked across the full range. The experiment is repeated at $\kappa \in \{0.03, 0.1, 0.3, 1, 3\}$, which moves the measured arbitrage share of volume across $89\%, 58\%, 25\%, 13\%, 16\%$ — from an over-provisioned pool dominated by arbitrage to a retail-flooded flagship. Each operating point is a full matrix and is never pooled with another. Table I reads the five levels as kinds of pools: κ is daily organic demand in units of the pool's own TVL, and the informed-heavy points are not stress tests — estimates of the informed share for non-primary venues routinely fall in the 40–70% range, so $\kappa \leq 0.1$ arguably describes the median DEX pool rather than an extreme. Figure 1 shows the construction at a glance: one measured intraday shape, five κ heights, each height a recognizable kind of pool with its measured flow mix.

C. Design and Statistics

One operating point is a full factorial over 9 policy configurations (five hook policies — `BAHook`, `DAHook`, `ABHook`, `MEVChargeHook`, and a volatility hook driven by an EMA of per-minute price variance — and static baselines at 5/30/60/100 bps) $\times$ 3 pairs $\times$ 72 windows $\times$ 3 gas scenarios — 5,832 cells; five operating points give 29,160 cells. Figure 2 illustrates the hooks' characters on one stormy example day before any aggregation: the directional ratchets split — the side the market leaves gets dearer, the opposite one cheaper — the constant-sum hook barely re-aims its budget, the volatility hook glides to the fee ceiling as variance arrives, and the size-surcharge hook alternates between its base and capped rate. The LP outcome of a cell is $\text{net} = R_f - IL = -L_{\text{net}}$ of the main paper's fee-adjusted loss, Eq. (M4): the LP's position and accrued fees against simply holding the deposit, all marked at end-of-window prices.[2]

Every comparison is *paired*: a policy's window result is differenced against the pre-declared 30 bps static baseline on the same window, pair, and gas price, removing the day-to-day variation that dwarfs policy effects. Effects are reported as the median paired difference with a 95% percentile-bootstrap confidence interval (10,000 seeded resamples); a comparison counts as a win or loss only if the interval excludes zero. Headline numbers aggregate cell-first (one median per pair$\times$regime$\times$gas cell, then across cells), so three gas scenarios of one day are never treated as independent observations.[3] A post-hoc comparison against the best static fee in the box (60 bps) is reported and labeled exploratory.

TABLE II
MEDIAN PAIRED ADVANTAGE OVER THE 30 BPS STATIC BASELINE AT THE REFERENCE POINT (κ=1): MEDIAN ACROSS THE 18 VOLATILE-PAIR STRATA, 95% BOOTSTRAP CI. NEGATIVE = WORSE THAN THE BASELINE.

Policy	**advantage vs 30 bps, USDT/window**
Static 60 bps	-2 $[-186, +266]$
`BAHook`	-100 $[-453, +350]$
`DAHook`	-235 $[-381, -104]$
`ABHook`	-460 $[-630, -288]$
Volatility hook	-752 $[-1069, -280]$
`MEVChargeHook`	-913 $[-1659, -728]$

D. Results

1) No unconditional winner: Table II gives the reference-point headline: no evaluated dynamic policy beats the static 30 bps baseline; four lose decisively and `BAHook` is indistinguishable from it. The same holds at every other operating point — pooled across regimes and gas scenarios, no hook's interval clears zero at any κ (Fig. 5, companion pooled figure in the artifact). The static fee curve itself has an interior optimum: mean outcomes of $1.6\text{k}/11.9\text{k}/11.9\text{k}/7.9\text{k}$ USDT per window at 5/30/60/100 bps, a 30–60 bps plateau consistent with the model's $\arg\max_f f e^{-\lambda|r(f)|} \approx 2/\lambda \approx 43$ bps —

[2]The artifact additionally reports a trade-time valuation (each trade marked at the price of its own moment); every finding below is unchanged under it.

[3]The analysis CSVs also carry classical Wilcoxon signed-rank tests with Benjamini–Hochberg correction across the $\sim$300 comparisons; they agree with the intervals in $\sim$96% of cases and are not used in the text.

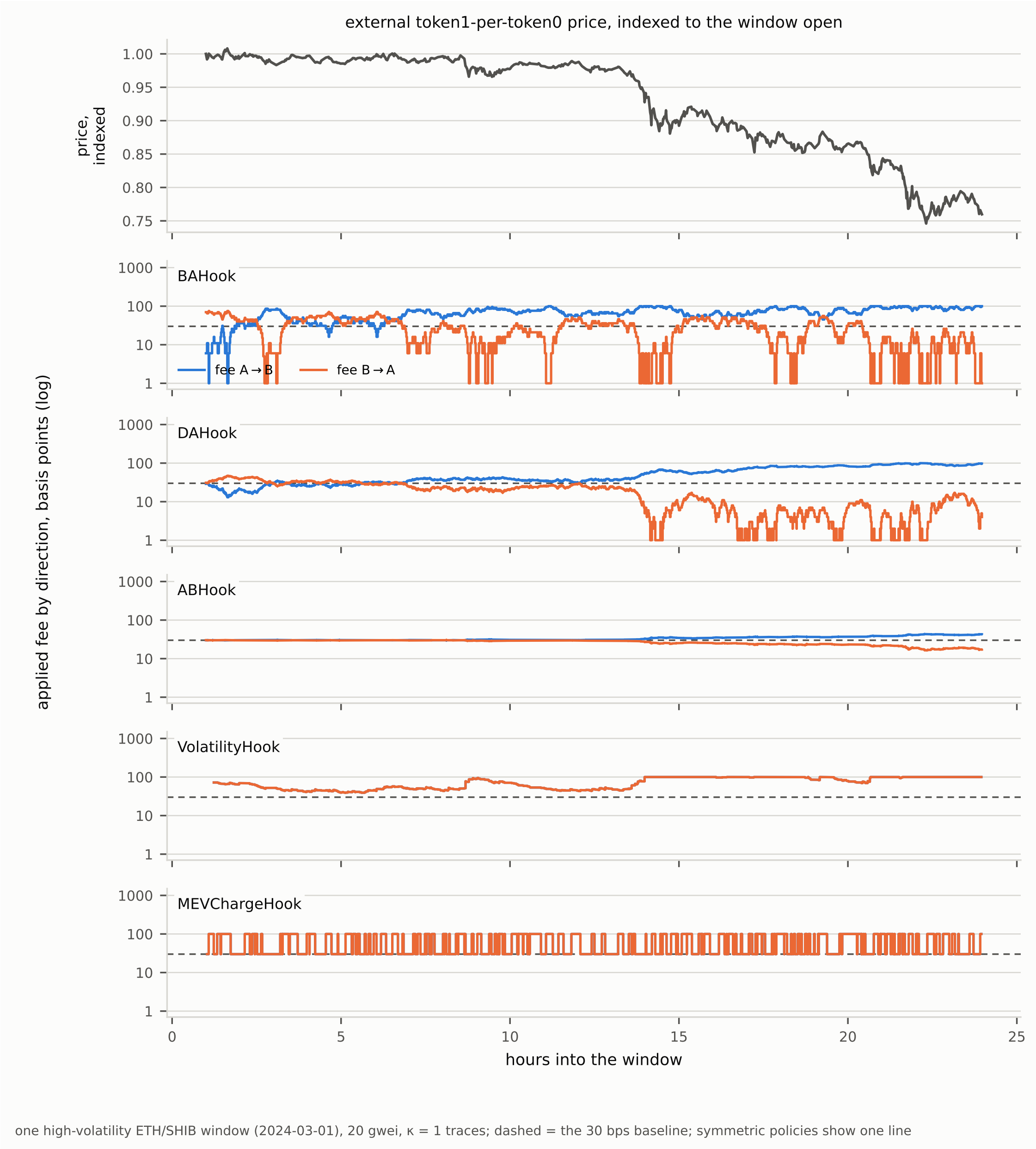

Fig. 2. What each hook actually charged through one high-volatility ETH/SHIB day (2024-03-01, 20 gwei, κ=1; log fee axis shared across panels; dashed = the 30 bps baseline). Each mark is the fee of an executed swap, held forward; for the directional hooks the two directions' fees interleave as swap direction alternates, which produces the sawtooth.

with the caveat that this optimum's location follows from the assumed participation function.

2) A fee policy prices flow; it does not protect: Decomposing the paired differences, the impermanent-loss term is policy-invariant (single-digit USDT against fee-income effects in the hundreds): arbitrage pins the end-of-window pool price to the external price under every policy, so adaptation cannot reduce adverse selection in this model — it can only price the flow, and it prices it slightly worse than a well-chosen constant while paying its own gas premium on top.

3) Where adaptation does pay: Crossing the two experimental axes reveals a robust conditional structure (Fig. 3). At κ=1, `BAHook` beats the baseline in high-volatility windows at 5 gwei by $+1{,}556$ $[+894, +2{,}501]$ USDT per window and in mid-volatility windows at 5 gwei by $+367$ $[+115, +571]$; at 80 gwei every policy loses or ties. The corner replicates at every operating point with a clean hand-over of ownership (Fig. 5): at retail-dominated mixes ($\kappa \geq 0.3$) it belongs to `BAHook`; at informed-heavy mixes ($\kappa \leq 0.1$) to the volatility hook and `DAHook` ($+308$ $[+32, +742]$ and $+246$ $[+59, +563]$ at κ=0.1, Fig. 4); and at the arbitrage-dominated extreme (κ=0.03) the volatility hook wins storms at *all three* gas prices — the retail-gas channel is irrelevant when there is hardly any

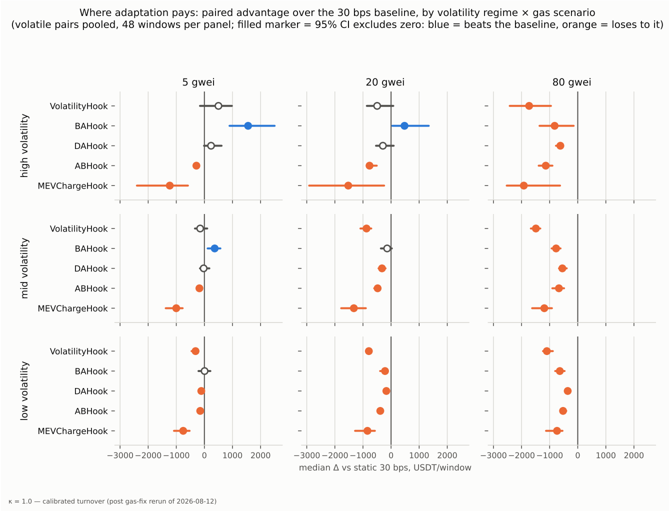

Fig. 3. Conditional structure at κ=1: median paired advantage over the 30 bps baseline by volatility regime × gas scenario (volatile pairs pooled, 48 windows per panel; filled marker = 95% CI excludes zero; blue wins, orange loses).

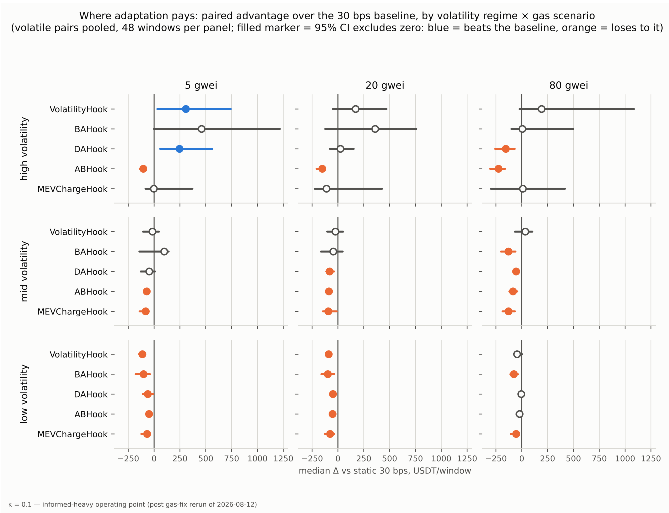

Fig. 4. The same decomposition at the informed-heavy point (κ=0.1, arbitrage 58% of volume): the storm–cheap-gas corner strengthens and changes owners.

retail. Read relatively (artifact figures), every policy improves monotonically with volatility at every gas level and degrades monotonically with gas in every regime.

4) Gas is a first-order term: The mechanism behind the gas gradient is the policies' own overhead: oracle-reading hooks cost 108–124k gas per swap against the static hook's 76k, retail prices that difference into Eq. (2), and at 80 gwei even the static hook's ∼\$19 per swap is ∼47 bps of a typical \$4,000 retail trade — transaction costs rival the fee box itself. Consistently, the informed share of volume rises with gas (12.4% to 13.7% across 5→80 gwei at κ=1): small retail is priced out first.

5) Summary: None of the evaluated dynamic policies outperforms a well-chosen static fee unconditionally. A reproducible conditional advantage exists in high-volatility windows at low gas prices; which policy carries it depends on the informed share of flow, and the advantage is consumed by the policies' own gas premium as gas rises. The framework's contribution is that this conditional structure is measurable at all — paired, reproducible, and decomposed along pre-declared axes.

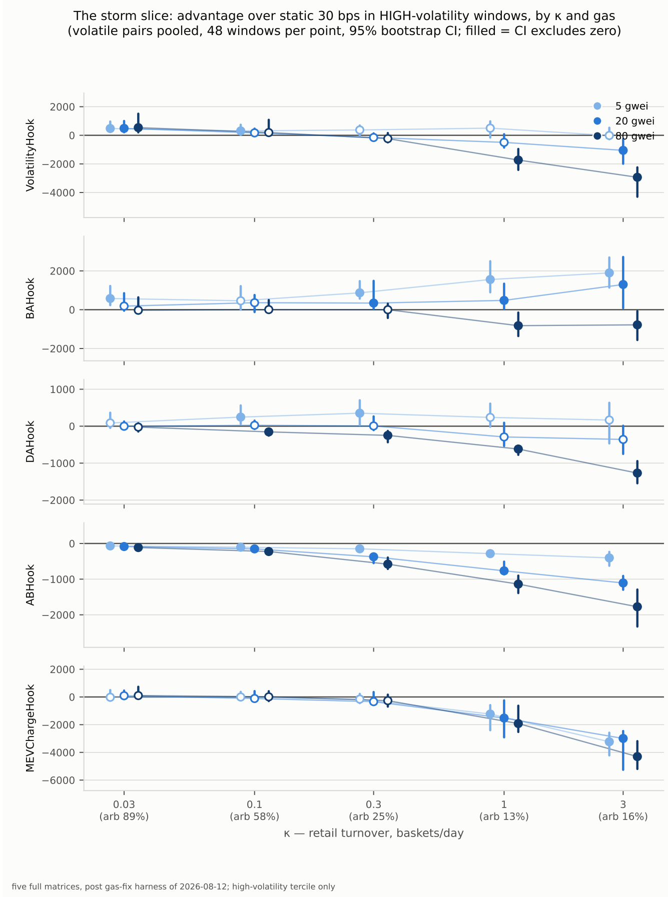

Fig. 5. The storm slice across the retail scale: advantage over the baseline in high-volatility windows, by κ and gas scenario.

E. Threats to Validity

The declared model limits: retail has no strategy and no correlated demand; one arbitrageur acts once per candle with perfect information and no mempool competition; liquidity is passive and full-range; mock feeds validate integration, not live oracle behavior; the retail scale κ is an assumption swept over two orders of magnitude rather than calibrated to real pools; retail lumpiness is fixed at one trade per direction per minute; results cover one year, three pairs, one venue's data, and a single RNG seed (a five-seed robustness appendix is defined in the artifact). The conditional corner finding was not pre-registered and is reported as observed structure; the unconditional findings and the failed-and-reported gas prediction follow the pre-registered protocol.

APPENDIX
DERIVATION OF THE ARBITRAGE OPTIMUM

Within one tick range a Uniswap v3/v4 pool with liquidity L obeys the constant-product amount relations in sqrt-price coordinates: moving the pool price from $\sqrt{p} = s_1$ to s_2 exchanges

$$\Delta y = L\,(s_2 - s_1), \qquad \Delta x = L\left(\frac{1}{s_1} - \frac{1}{s_2}\right), \tag{6}$$

and the LP fee f ($\gamma = 1 - f$) is charged on the input amount before it enters the curve.

1) Band edges: Selling token0 (zero-for-one) at pool sqrt-price s realizes a marginal price γs^2 in token1 per token0; the trade is profitable against the external price P while $\gamma s^2 > P$, i.e. while $s > \sqrt{P/\gamma} = t_{0\to1}$. Buying token0 costs s^2/γ marginally and is profitable while $s^2/\gamma < P$, i.e. while $s < \sqrt{P\gamma} = t_{1\to0}$. Hence the no-arbitrage band $[P\gamma,\, P/\gamma]$ on the pool price, and the profit-maximizing trade stops exactly at the near edge: the integrand of marginal profit changes sign there, so profit is unimodal in trade size with its maximum at the edge.

2) Sell side: Moving the pool from s down to $t = t_{0\to1}$ requires a net token0 input $\Delta x = L(1/t - 1/s)$, for which the trader pays the grossed-up $\Delta x/\gamma$, and yields $\Delta y = L(s-t)$ of token1. Valued at the external price $P = \gamma t^2$,

$$\begin{aligned}\Pi_{0\to1} &= L(s-t) - \frac{P}{\gamma}\,L\left(\frac{1}{t} - \frac{1}{s}\right) = L(s-t) - L\,t^2\,\frac{s-t}{ts}\\ &= L(s-t)\left(1 - \frac{t}{s}\right) = \frac{L\,(s-t)^2}{s}.\end{aligned} \tag{7}$$

3) Buy side: Moving the pool from s up to $t = t_{1\to0}$ requires net token1 input $L(t-s)$, paid as $L(t-s)/\gamma$, and yields $\Delta x = L(1/s - 1/t)$ of token0 worth $P\,\Delta x$. With $P = t^2/\gamma$,

$$\begin{aligned}\Pi_{1\to0} &= \frac{t^2}{\gamma}\,L\,\frac{t-s}{ts} - \frac{L(t-s)}{\gamma} = \frac{L(t-s)}{\gamma}\left(\frac{t}{s} - 1\right)\\ &= \frac{L\,(t-s)^2}{\gamma\,s}.\end{aligned} \tag{8}$$

The extra $1/\gamma$ relative to Eq. (7) arises because the fee is charged in token1 — the numeraire the profit is measured in — so, unlike the sell side, it does not cancel against the external valuation of the input.

4) Gas: Gas adds a size-independent cost $g\,p_{\text{gas}}$, which shifts the profit curve down without moving its maximum; checking $\Pi^* > g\,p_{\text{gas}}$ after the size optimization is therefore equivalent to including gas inside it (the arbitrageur cannot split a trade across transactions in this model). For a size-dependent fee the closed form is replaced by a ternary search, valid because an input fee non-decreasing in size preserves unimodality.

REFERENCES

[1] Irina Lebedeva, Dmitrii Umnov, Yury Yanovich, Ignat Melnikov, and George Ovchinnikov. Dynamic fee for reducing impermanent loss in decentralized exchanges, 2025.